Comment on
“Framework for liquid crystal based particle models”

Intellectual Property Risks and Platform Deficiencies in the OpenWave Platform

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Abstract

This Comment reviews the codebase architecture and licensing terms of the OpenWave simulation platform, which serves as the numerical implementation environment for the liquid crystal particle model in Ref. (Duda 2021). The codebase consists of fragmented, redundant, and structurally incomplete sandbox files that do not execute without custom, unreleased internal dependencies. Furthermore, significant intellectual property risks exist for researchers: the contributor guidelines and the Apache License 2.0 permit commercial platforms to harvest and exploit user-submitted models without royalty or attribution, while the classification of artificial intelligence agents as first-class contributors establishes potential developer lock-in.

Section 1: Codebase Quality and Structural Incompleteness

The OpenWave platform is presented as a professional, unified testing ground for emergent physics. However, a review of the codebase reveals that the files are poorly organized, highly coupled, and incomplete: * The scripts are a disorganized collection of local, fragmented sandbox folders (e.g. `sandbox_v2`, `sandbox_v6`, `sandbox_v8`, `sandbox_vn`) that contain duplicated logic and obsolete algorithms. * The codebase scripts are highly coupled to local, undocumented repository configurations. For example, even when a script is structurally complete (such as `m5_8_2q_delta_scaling.py`), it cannot be run out of the box because it imports custom modules (like `openwave.xperiments`) that are not packaged or distributed on standard package indexers like PyPI. * The repository is not structured as a standard Python package, requiring manual manipulation of `sys.path` to resolve imports, which often fail due to missing files and directories in public distributions.

Section 2: Intellectual Property and Licensing Risks

OpenWave operates under the **Apache License, Version 2.0**. While this is a standard open-source license, its application to advanced research models uploaded to a centralized platform carries significant risks for independent researchers: * **Commercial Exploitation Without Compensation**: The Apache 2.0 license explicitly grants anyone (including corporate entities and the platform owners) the right to use, modify, and distribute the code for commercial purposes without royalty or compensation to the original authors. * **Patent Grants**: By contributing code to the platform, authors automatically grant a perpetual, worldwide, non-exclusive, no-charge, royalty-free, irrevocable patent license to all users of the software. If a researcher's model contains novel mathematical methods or device concepts, they effectively waive their rights to patent protection. * **Lack of Contributor Protections**: OpenWave lacks a structured Contributor License Agreement (CLA) that protects the intellectual property rights of human authors or ensures they retain exclusive rights to commercial derivatives.

Section 3: AI Agent Harvesting and Developer Lock-In

OpenWave's developer documentation highlights a novel approach: treating AI agents as **"first-class contributors"** alongside human researchers. This represents a structural risk to the community: * The platform is designed to automate the integration and testing of models by leveraging AI agents to sweep research repositories, extract models, and run them against benchmarks. * This creates an automated system for harvesting research ideas and converting them into platform-owned, open-source code under the Apache 2.0 license, stripping the original human researchers of control over their work. * **Warning to the Community**: Researchers are strongly cautioned against uploading proprietary models or working within the OpenWave environment. Submitting to this platform risks losing the rights to original work, which can then be freely monetized by the platform owners or other commercial entities.

Section 4: Conclusion

The OpenWave simulation platform introduces substantial code-quality, licensing, and intellectual property risks for scientific developers. The presence of incomplete sandbox folders, unreleased internal dependencies, and permissive Apache 2.0 terms allows commercial entities to exploit contributed physics models without compensation, while the platform's AI agent ingestion framework threatens to automate the harvesting of developer models.

References

Duda, Jarek. 2021. "Framework for Liquid Crystal Based Particle Models." *arXiv Preprint arXiv:2108.07896*. https://arxiv.org/abs/2108.07896.